Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/07/2010
PI Name:	Li, Chuan-Yuan Ph.D.		
Project Title:	A mechanistic investigation of space radiation-induced carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chuan.li@duke.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	919 613 8754
Organization Name:	Duke University		
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City:	Durham	**State:**	NC
Zip Code:	27710	**Congressional District:**	4
Comments:	PI moved to Duke University in December 2011. Formerly at University of Colorado Denver (Ed., 2/8/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	06/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2012 per NSSC/A. Chu/ARC (Ed., 3/23/2011) NOTE: Received NCE until 5/31/2011 (previous end date 5/31/2010), per A. Chu/ARC (6/2010)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Li, Fang (University of Colorado Denver)		
Grant/Contract No.:	NNX09AH19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of major concerns for manned space missions of NASA is exposure to galactic cosmic rays (GCRs) or highly charged energetic (HZE) particles, which carries distinct health risks. The major goal of the NASA Bioastroanautics Roadmap and NASA ground-based studies in radiation biology is to assess potential risks of human exposure to HZE particles and to generate knowledge that can be used to mitigate the health risks of HZE particle exposure eventually. In this project, we will two specific aims to study HZE particle-induced mutagenesis and carcinogenesis in mammalian cells. These are: 1. To determine the potential interactions of reactive oxygen/nitrogen species and apoptosis in regulating HZE radiation-induced mutagenesis in mammalian cells. 2) to determine the roles of program cell death in HZE radiation induced mutagenesis/carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will have the following potential benefit for life on earth: 1) we will develop a state-of-art techniques to monitor radiation induced DNA damage, which will facility a better understanding radiation induced carcinogenesis in humans. 2) Our study may lead to fundamental insights into how cells deal with DNA damage. 3) we hope to achieve novel, mechanistic insights into the carcinogenic risks of radiation, which is universally present on earth.
Task Progress:	1) We have established a novel approach to monitor DNA double strand breaks through quantifying H2AX-MDC1 interaction in irradiated mammalian cells. The monitoring could be carried out in a non-invasive manner. In addition, it could be quantified in a serial manner, making it possible to monitor the status of DNA damage and repair in irradiated cells or tissues over a long period of time. 2) We have made three runs of experiments (Spring, Summer and Fall) in 2009 with (0.6 GeV) Fe ions at NSRL facility located in BNL. All three experiments went through smoothly. 3) With these runs we have obtained the following preliminary results, with more repetitions necessary in the planning in the coming NSRL runs: • We have established a cell line H322-H2AX reporter, with our H2AX-MDC1 reporter. The cell line was derived from human lung cancer. • We have used the H322-H2AX reporter for irradiation at NSRL runs. • Because of limitation of detection (the lack of a luciferase imaging instrument), we were not able to detect the earlier time points. However, because there are already much published data at the earlier time points, we focused our attention at later time points. One striking feature of our data is the persistent levels of H2AX activation after Fe ion exposure. This perhaps reflect the difficulties in repairing damage induced heavy ions, which could may profound implications in space radiation induced mutagenesis and carcinogenesis.
Bibliography Type:	Description: (Last Updated: 10/30/2019)
Articles in Peer-reviewed Journals	Li F, Huang Q, Chen J, Peng Y, Roop DR, Bedford JS, Li CY. Sci Signal. 2010 Feb 23;3(110):ra13. "Apoptotic cells activate the “Phoenix Rising” pathway to promote wound healing and tissue regeneration." Science Signaling, 2010 Feb 23;3(110):ra13. PMID: 20179271 http://dx.doi.org/10.1126/scisignal.2000634 , Feb-2010
Journal/Magazine covers	Li F, Huang Q, Chen J, Peng Y, Roop DR, Bedford JS, Li CY. "This article, 'Apoptotic cells activate the 'Phoenix Rising' pathway to promote wound healing and tissue regeneration, ' is featured on the journal cover, with an illustration depicting a phoenix rising from a tissue injury site." Science Signalling 2010 Feb 23;3(110): cover. http://stke.sciencemag.org/content/vol3/issue110/ , Feb-2010